

The University of Nottingham

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2021-2022

Mathematics for Computer Scientists 2

Time allowed: X Hours and Y Minutes

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced

«Answer Rubric»

«Calculator Rubric»

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

ADDITIONAL MATERIAL: «List Additional/Supplementary Materials MUST be specified in this section of the paper Handouts, MCQ, number of answer books x , graph paper, statistical tables (Neaves, SI, E&E), statutes, Books allowed, None».

INFORMATION FOR INVIGILATORS: «to include any announcements or to indicate if the paper can be taken away at the end of the examination / None».

SECTION A

QUESTION 1**1. Set Theory**

Calculate $A \times B$ where

$$A = \{a, b, c\}$$

and

$$B = \{x \in \mathbb{Z} \mid x^2 - 2x - 8 = 0\}.$$

[< < > > Marks]

QUESTION 2**2. Matrices**

Let us consider the following matrix

$$\begin{pmatrix} 1 & -2 & 0 \\ 2 & 2 & 6 \\ 0 & 1 & 1 \end{pmatrix}$$

- Determine the rank of the matrix

[< < > > Marks]

- Invert the matrix if possible. If it is impossible, justify your answer.

[< < > > Marks]

End of Question 2: Total < < > > marks

QUESTION 3**3. Gaussian elimination**

Let us consider the following system of linear equations

$$\begin{cases} x - y + z = 1 \\ x + y = 4 \\ 2x + 2y + 2z = 9 \end{cases}.$$

- Apply, if possible, the Gaussian elimination to find the equivalent triangular matrix/system

[< < > > Marks]

- Use the Gaussian elimination for performing the LU factorization

[< < > > Marks]

- With the detected triangular system find the solution of the system

[< < > > Marks]

End of Question 3: Total < < > > marks

QUESTION 4**4. Vector Spaces**

Let us consider the vector space $(U, +, \cdot)$ where

$$U = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \begin{cases} x + 2y - z = 0 \\ 2x + 4y - 2z = 0 \\ 6x + 12y - 6z = 0 \end{cases} \right\}.$$

- Determine a basis of the vector space

[< < > > Marks]

- Determine the dimension of the vector space

[< < > > Marks]

- Let $(V, +, \cdot)$ be a vector space whose basis $B_v = \{(0, 0, 0)\}$. Verify the Grassmann's formula for these two vector spaces.

[< < > > Marks]

- Provide a geometric interpretation of what Grassmann's formula states for U and V

[< < > > Marks]

End of Question 4: Total < < > > marks

SECTION B

QUESTION 1**1. Matrices**

Let $\alpha, \beta \in \mathbb{R}$. Let us consider the following matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \alpha a_{11} + \beta a_{12} \\ a_{21} & a_{22} & \alpha a_{21} + \beta a_{22} \\ a_{31} & a_{32} & \alpha a_{31} + \beta a_{32} \end{pmatrix}$$

with $a_{ij} \in \mathbb{R}$ and $a_{ij} \neq 0, \forall i, j$. Let us now consider a matrix $\mathbf{B}$ such that

$$\mathbf{AB} = \mathbf{O}$$

with $\mathbf{O}$ null matrix.

- Determine the possible rank values of $\mathbb{B}$. Justify your answer and use the relevant theory.

[< < > > Marks]

- Can it ever happen that $\mathbf{B}$ is non-singular? Justify your answer.

[< < > > Marks]

- Can it ever happen that **B** is the null matrix? Justify your answer.

[< < > > Marks]

End of Question 1: Total < < > > marks

QUESTION 2

2. Vector Spaces/Linear Mappings

Let $(U, +, \cdot)$ be a vector subspace of $(\mathbb{R}^n, +, \cdot)$ with $\dim(U) < n$. Let $f : U \rightarrow \mathbb{R}^n$ be a linear mapping.

- Assess whether or not the mapping can be injective. Justify your answer by using the relevant theorems.

[< < > > Marks]

- Assess whether or not the mapping can be surjective. Justify your answer by using the relevant theorems.

[< < > > Marks]

- Assess whether or not the mapping can be invertible. Justify your answer by using the relevant theorems.

[< < > > Marks]

End of Question 2: Total < < > > marks